RAJARAPU VENKATESWARLU

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PROFESSIONAL SUMMARY

Final-year B.Tech Data Science student (expected May 2027) with Python and SQL skills and hands-on experience in machine learning, computer vision, and FastAPI backend development. Completed two AI/ML internships and projects spanning YOLOv8 defect detection and explainable quality-risk prediction. Seeking entry-level Data Science, Machine Learning, or AI Engineer roles.

TECHNICAL SKILLS

Languages: Python, SQL
Machine Learning: Scikit-learn, XGBoost, LightGBM, Random Forest, SMOTE, Feature Engineering, Classification, Model Evaluation (Precision, Recall, F1-score, ROC-AUC, PR-AUC), SHAP
Data Analysis: Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Data Preprocessing, Microsoft Excel
Deep Learning and Computer Vision: YOLOv8, ResNet18, OpenCV, TensorFlow
Backend and Web: FastAPI, Flask, REST APIs, JWT Authentication, React, HTML, CSS, Tailwind CSS
Databases: PostgreSQL, MySQL, SQLite
Tools and Deployment: Git, GitHub, Docker, Render
CS Fundamentals: Data Structures and Algorithms, Object-Oriented Programming, DBMS

EXPERIENCE

Infosys Springboard | AI/ML Virtual Intern (Virtual Internship 7.0, Batch 2) Jul 2026 - Sep 2026

- Engineered VisionInspect AI end to end as a solo project, taking it from model development through backend, frontend, and deployment.
- Implemented JWT authentication in a FastAPI backend, with role-based access control for two user roles.
- Designed a 12-table PostgreSQL schema to support the application's data storage.

APSCHE SmartBridge | AI/ML Intern Apr 2026 - May 2026

- Prepared a 614-record loan dataset by cleaning data, encoding categorical features, and applying feature scaling; addressed class imbalance with SMOTE.
- Compared Decision Tree, Random Forest, KNN, and XGBoost classifiers for the Smart Lender loan-eligibility project; XGBoost performed best with 81.1% test accuracy.
- Integrated the selected model into a Flask web application (HTML, CSS, MySQL) for loan-eligibility prediction.

PROJECTS

VisionInspect AI - Manufacturing Defect Detection and Quality Inspection System

- Applied YOLOv8 for defect detection and localization and ResNet18 for pass/fail classification, with OpenCV image processing and a contour-based fallback for image analysis.
- Developed React interfaces for image inspection, inspection history, dashboards, and reports, backed by FastAPI REST APIs for model inference.
- Containerized the application with Docker and deployed it on Render, connecting the frontend, APIs, model inference, PostgreSQL database, and image storage.

NSQ-Radar - Pharmaceutical Quality Risk Prediction Dashboard

- Processed a synthetic 1,000,000-record dataset, engineered features from inspection, manufacturer, and quality attributes, addressed class imbalance, and compared Random Forest, LightGBM, and XGBoost models to predict Not of Standard Quality (NSQ) risk.
- Evaluated the XGBoost model on the test set: 76.71% recall on the NSQ class, 0.8680 ROC-AUC, and 0.5575 PR-AUC.
- Created a React and FastAPI dashboard (PostgreSQL, Tailwind CSS) that ranks potentially high-risk products and uses SHAP to explain individual predictions.

EDUCATION

B.Tech, Computer Science and Engineering (Data Science) | NRI Institute of Technology, Guntur
CGPA: 8.5/10 | Expected Graduation: May 2027

CERTIFICATIONS

- Oracle Certified Foundations Associate - Agentic AI | Oracle University | Aug 2026
- Career Essentials in Generative AI | Microsoft and LinkedIn Learning | Aug 2026
- Artificial Intelligence Fundamentals | IBM SkillsBuild | Aug 2026
- Predictive Analytics using Python and Excel | AmyPro Technologies Pvt. Ltd. | Jul 2026